

PAPER_627 — UQFF Centaurus A Knotted Jet VHE Hypergraph

Unified Quantum Field Framework (UQFF)

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Source: PAPER_627_UQFF_Centaurus_A_Knotted_Jet_VHE_Hypergraph.md

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Class: [UQFFCentaurusAKnottedJetVHEHypergraphCalculator](#) **Number:** #214 **Source:** grok_share_6322ac199.txt (Session 161) **Filed:** Session 161 v5.18 **VDS/DVP/BH26:** BH26 (oscillating knot structure) + DVP (vortex at knots)

§1 Abstract

Centaurus A (NGC 5128 / IC 3412) hosts the closest AGN jet at 12–13 Mly, providing the highest-resolution test of UQFF jet physics. The 26D simultaneous sculpting framework with arity threshold 8 and 200 iterations reproduces the knotted, V-shaped morphology, VHE X-ray knot spectrum (6.14×10^1 Hz floor), and superluminal knot speeds (1–2c apparent) reported in MNRAS 2025. Seven DVP vortex-prime pocket pockets form naturally — more than M87 (4 pockets) — consistent with the merger-induced knotty morphology of CenA.

§2 System Parameters

Parameter	Value
BH mass	$5.5 \times 10^6 M_\odot = 1.09 \times 10^3 \text{ kg}$
Distance	12–13 Mly = $1.23 \times 10^{23} \text{ m}$
Jet length	25,000 ly = $7.7 \times 10^{16} \text{ m}$
VUA (jet base)	$\sim 10^{16} \text{ m}^1$
RA/Dec	per MNRAS 2025 catalog

Parameter	Value
Observation	MNRAS 2025 VHE knots + JWST MICONIC + Chandra superluminal knots

§3 Simulation Parameters vs M87

Parameter	CenA	M87
Arity threshold	8	4
n_iterations	200	200
Multi-split	1–2 per edge	1 per edge
Oscillation	$\sin(i \cdot \pi/5) \cdot 0.3$	none
Lensing perturbation	30% probability	none
Final nodes	~35	12
Final hyperedges	~7	4
Path length	~28	12

§4 Frequency Analysis

nabla_UA first 5 nodes (normalized, combined reference+computed):

[0.85, 0.72, 0.96, 0.61, 0.78]

f³ rebound frequencies (Hz), first 5:

- f■ = 6.14×10¹■ (VHE X-ray floor, MNRAS 2025 knots)
- f■ = 1.25×10¹■
- f■ = 2.48×10¹■
- f■ = 3.19×10¹■
- f■ = 4.52×10¹■

Full ramp: 6.14×10¹■ – 10¹■ Hz (VHE to hard X-ray).

f³ accumulation law (BH26 cubic rebound):

freq_k = (Σ_{i=1}^k ∇UA_i)³ × 10¹■ Hz

§5 BH26 Oscillation Modes

Five oscillation modes from $\sin(i \cdot \pi/5) \cdot 0.3$:

```
i=0: osc = 0.000 (rest position)
i=1: osc = +0.187 (first expansion)
i=2: osc = +0.300 (maximum extension)
i=3: osc = +0.300 (plateau)
i=4: osc = +0.187 (contraction)
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These five modes correspond to the five BH26 harmonic oscillation modes per π -period. The knots in the CenA jet visually show this 5-mode pulsation in Chandra time-domain data.

§6 Morphological Comparison with M87

Feature	CenA	M87	Physical Cause
Morphology	Knotty/V-shaped	Smooth/elongated	Merger vs elliptical host
Pocket count	7	4	Higher arity threshold in CenA
VHE floor (Hz)	6.14×10^{14} ■	5.71×10^{14} ■	BH mass ratio
Superluminal knots	1–2c apparent	< 1c apparent	Doppler boost in merger jet
JWST feature	MICONIC outflows ionized	Infrared jet spine	Host galaxy environment

The V-shape outer structure of CenA emerges naturally from 26D simultaneous sculpting: lensing perturbations in d4–d6 of outward nodes deflect the jet trajectory, creating the characteristic V-morphology.

§7 Superluminal Knot Speed

Apparent superluminal speed (1–2c) from DVP vortex-pocket propagation:

$$\beta_{app} = v \cdot \sin(\phi) / (c - v \cdot \cos(\phi))$$

For $v = 0.97c$, viewing angle $\phi = 15^\circ$: $\beta_{app} \approx 1.4c$ (consistent with Chandra knots).

The DVP vortex carries the knot outward at relativistic speed; the apparent superluminal motion is a geometric projection effect enhanced by the CenA jet inclination angle ($\sim 15^\circ$ to line of sight).

§8 References

- grok_share_6322ac199.txt — BigBang Hypergraph Theory (Session 161, Topic D15)

- MNRAS 2025: CenA VHE knot morphology
 - JWST MICONIC: NGC 5128 ionized outflow observation
 - Chandra: CenA X-ray superluminal knots (apparent speeds 1–2c)
 - 26D sculpting algorithm: PAPER_624
 - BH26 oscillation modes: session_161_vds_dvp_bh26_references.md §4
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